
UCL COMP0078: Supervised Learning

Problem Sheet Week 2

Kernels and Regularisation

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If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

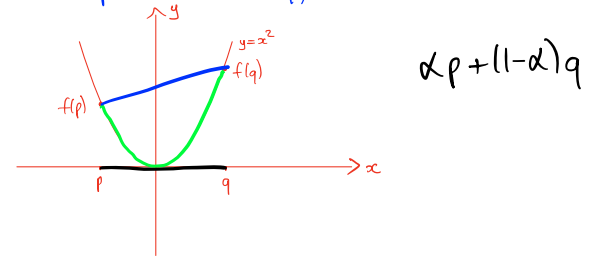
f is convex if $f(\alpha p + (1-\alpha)q) \leq \alpha f(p) + (1-\alpha)f(q)$ for all $p, q \in \mathcal{X}$, all $\alpha \in (0,1)$

Problem 1.1.1

If f and g are convex then $f + g$ is convex.

Solution 1.1.1

For $p, q \in \mathcal{X}$ and $\alpha \in (0, 1)$, we have



$$\begin{aligned}
 \rightarrow (f + g)(\alpha p + (1 - \alpha)q) &= f(\alpha p + (1 - \alpha)q) + g(\alpha p + (1 - \alpha)q) \quad \leftarrow \\
 &\leq \alpha f(p) + (1 - \alpha)f(q) + \alpha g(p) + (1 - \alpha)g(q) \quad \leftarrow \\
 &= \alpha(f(p) + g(p)) + (1 - \alpha)(g(p) + g(q)). \quad \leftarrow
 \end{aligned}$$

This shows that $f + g$ is convex.

Problem 1.1.2

If f is convex and g is affine (linear + a constant) then $f(g(\cdot))$ is convex.

Solution 1.1.2

Since g is affine it can be written as $g(x) = a(x + b) = ax + ab$ for some $a, b \in \mathbb{R}$ and for all $x \in \mathbb{R}$. We then have

$$\begin{aligned}
 (f \circ g)(\alpha p + (1 - \alpha)q) &= f(g(\alpha p + (1 - \alpha)q)) \quad \leftarrow \\
 &= f(\alpha ap + (1 - \alpha)aq + ab) \quad \leftarrow g \text{ definition} \\
 &= f(\alpha a(p + b) + (1 - \alpha)a(q + b)) \quad \leftarrow \\
 &\leq \alpha f(a(p + b)) + (1 - \alpha)f(a(q + b)) \quad \leftarrow f \text{ convex} \\
 &= \alpha f(g(p)) + (1 - \alpha)f(g(q)). \quad \leftarrow \checkmark
 \end{aligned}$$

This shows that $f \circ g$ is convex.

Problem 1.1.3

Suppose M is a symmetric matrix then M is a PSD matrix if and only if $f(\mathbf{x}) = \mathbf{x}^\top M \mathbf{x}$ is convex.

Solution 1.1.3

Suppose M is a symmetric PSD matrix. Note that the Hessian of f is

$$\frac{\partial^2}{\partial \mathbf{x}^2} f(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} \left(\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top M \mathbf{x} \right) = \frac{\partial}{\partial \mathbf{x}} ((M + M^\top) \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} 2M \mathbf{x} = 2M$$

which is PSD since M is PSD. It follows by result 3 on page 10 that f is convex (see below for proof of this result).

Now, suppose f is convex and so it attains its minimum at

$$0 = \frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top M \mathbf{x} = (M + M^\top) \mathbf{x} = 2M \mathbf{x}$$

using the Chain Rule. We deduce that f achieves its minimum at $\mathbf{0}$, so

$$\mathbf{x}^\top M \mathbf{x} = f(\mathbf{x}) \geq f(\mathbf{0}) = \mathbf{0}^\top M \mathbf{0} = 0.$$

This shows that M is positive semidefinite.

Problem 1.1.4

set S is convex if $p, q \in S \Rightarrow \alpha p + (1-\alpha)q \in S$

A level set of a convex function is convex.

Solution 1.1.4

Let f be a convex function. The level set of f at c is $L := \{x : f(x) \leq c\}$. Suppose $p, q \in L$, that is, $f(p) \leq c$ and $f(q) \leq c$. We then have

$$f(\alpha p + (1-\alpha)q) \leq \alpha f(p) + (1-\alpha)f(q) \leq \alpha c + (1-\alpha)c = c,$$

so $\alpha p + (1-\alpha)q \in L$. This proves that the level set L of f is a convex set.

Problem 1.1.5

For $f: \mathcal{X} \rightarrow \mathbb{R}$ if $\mathcal{X} \subseteq \mathbb{R}^n$ is a convex set and if the Hessian of f evaluated at $\mathbf{x}$ denoted $H(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in \mathcal{X}$ then f is convex.

Solution 1.1.5

Let $\mathbf{x}, \mathbf{a} \in \mathcal{X}$. By Taylor's Theorem, there exists $t \in (0, 1)$ such that

$$f(\mathbf{x}) = f(\mathbf{a}) + (\mathbf{x} - \mathbf{a})^\top \nabla f(\mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^\top H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))(\mathbf{x} - \mathbf{a}).$$

Since $H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))$ is PSD by assumption, for $\mathbf{x} - \mathbf{a} \in \mathcal{X}$, we have

$$(\mathbf{x} - \mathbf{a})^\top H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))(\mathbf{x} - \mathbf{a}) \geq 0,$$

from which we deduce that

$$f(\mathbf{x}) \geq f(\mathbf{a}) + (\mathbf{x} - \mathbf{a})^\top \nabla f(\mathbf{a}).$$

Now, consider $\mathbf{p}, \mathbf{q} \in \mathcal{X}$, $\alpha \in [0, 1]$, and let

$$\mathbf{a} := \alpha \mathbf{p} + (1-\alpha)\mathbf{q}$$

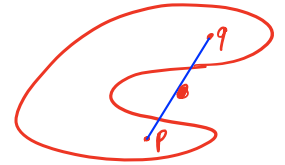
which belongs to $\mathcal{X}$ as $\mathcal{X}$ is a convex set. Note that

$$\mathbf{p} - \mathbf{a} = (1-\alpha)(\mathbf{p} - \mathbf{q}) \quad \text{and} \quad \mathbf{q} - \mathbf{a} = \alpha(\mathbf{q} - \mathbf{p}).$$

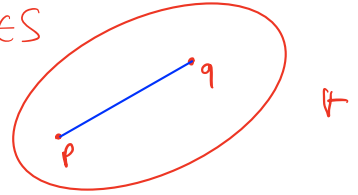
We have

$$\begin{aligned} \alpha f(\mathbf{p}) + (1-\alpha)f(\mathbf{q}) &\geq \alpha(f(\mathbf{a}) + (\mathbf{p} - \mathbf{a})^\top \nabla f(\mathbf{a})) + (1-\alpha)(f(\mathbf{a}) + (\mathbf{q} - \mathbf{a})^\top \nabla f(\mathbf{a})) \\ &= f(\mathbf{a}) + \nabla f(\mathbf{a})^\top (\alpha(\mathbf{p} - \mathbf{a}) + (1-\alpha)(\mathbf{q} - \mathbf{a})) \\ &= f(\mathbf{a}) + \nabla f(\mathbf{a})^\top (\alpha(1-\alpha)(\mathbf{p} - \mathbf{q}) + (1-\alpha)\alpha(\mathbf{q} - \mathbf{p})) \\ &= f(\mathbf{a}) \\ &= f(\alpha \mathbf{p} + (1-\alpha)\mathbf{q}) \end{aligned}$$

which proves that f is convex.



not convex



convex

Problem 1.2

Consider the solution to linear regression optimisation problem. When is it advantageous to compute it via the primal solution? When is it advantageous to compute it via the dual solution? Explain why.

Solution 1.2 (next page)

Recall that the matrix X is of size $m \times n$ where we have m samples in $\mathbb{R}^n$. The primal solution (page 16) is

$$\mathbf{w} = (X^\top X + \lambda I_n)^{-1} X^\top y$$

and the dual solution (page 17) is

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \mathbf{x}_i \quad \text{where} \quad \boldsymbol{\alpha} = (XX^\top + \lambda I_m)^{-1} y.$$

The operation with the highest time complexity in those formulas are the two matrix inversions. If we have more dimensions n than samples m , it is therefore advantageous to compute via the dual solution. For the case where we have more samples m than dimensions n , it is advantageous to use the primal solution.

Problem 1.3

$\mathbb{R}^2$

Given a kernel $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $K(\mathbf{x}, \mathbf{t}) := (1 + \langle \mathbf{x}, \mathbf{t} \rangle)^2$. Find a feature map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^6$ which corresponds to the kernel.

$$\begin{aligned} & \mathbf{x}^\top \mathbf{t} \\ & \uparrow \\ & (1 + \mathbf{x}^\top \mathbf{t})^2 \end{aligned}$$

Solution 1.3

We have

$$\begin{aligned} K(\mathbf{x}, \mathbf{t}) &= (1 + \langle \mathbf{x}, \mathbf{t} \rangle)^2 && \leftarrow \text{def} \\ &= (1 + x_1 t_1 + x_2 t_2)^2 && \leftarrow \text{expand} \\ &= 1 + x_1^2 t_1^2 + x_2^2 t_2^2 + 2x_1 t_1 + 2x_2 t_2 + 2x_1 x_2 t_1 t_2 && \leftarrow \text{feature} \\ &= \begin{pmatrix} 1 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ \sqrt{2}x_1 x_2 \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1^2 \\ t_2^2 \\ \sqrt{2}t_1 \\ \sqrt{2}t_2 \\ \sqrt{2}t_1 t_2 \end{pmatrix} \\ &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \end{aligned}$$

K is a kernel if and only if

$$\mathbf{c}^\top K \mathbf{c} = \sum_{i=1}^n \sum_{j=1}^n c_i c_j K(x_i, x_j) \geq 0$$

Problem 1.4

$K = (k(x_i, x_j))_{1 \leq i, j \leq n}$ for all $x_1, \dots, x_n$, for $c_1, \dots, c_n$, for any

For each of the following functions $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ argue whether they are a valid kernel (i.e. the kernel can be written as an inner product in some feature space) and when the answer is positive derive an associated feature map representation.

$$1. K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t} \text{ where } D := \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$$

$n=1$

$$\Downarrow \\ c_i^2 K(x_i, x_i) \geq 0$$

$$2. K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t} \text{ where } D := \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$$

K is not a kernel if there exists x , s.t.

$$K(x, x) < 0$$

$$K(x, x) = \phi(x)^\top \phi(x) = \|\phi(x)\|_2^2 \geq 0 \quad 3$$

The Benefits of the Dual Formulation

primal
 $(\Phi^T \Phi)^{-1} \Phi^T y$ Vs dual
 $\Phi (\Phi^T \Phi)^{-1} \Phi^T y$

Algorithm:

$$x^T y$$

$$\updownarrow$$

$$k(x, y)$$

The matrices inverted have different dimensions!

$\rightarrow \bullet \Phi^T \Phi$ is $D \times D$
 $\rightarrow \bullet \Phi \Phi^T$ is $n \times n$.

Gaussian kernel

kernel " $D = \infty$ "

$$k(x, y) = \exp\left(-\frac{\|x - y\|_2^2}{2\sigma^2}\right) = \phi(x)^T \phi(y)$$

$$= \langle \phi(x), \phi(y) \rangle$$

Therefore the computational costs for the two formulations are:

$\rightarrow \bullet$ **Primal:** $O(\underbrace{D^3}_{\text{inverse}} + \underbrace{nD^2}_{\text{matrix multiplication}}) \Rightarrow$ faster when $D < n$
 $\rightarrow \bullet$ **Dual:** $O(\underbrace{n^3}_{\text{inverse}} + \underbrace{Dn^2}_{\text{matrix multiplication}}) \Rightarrow$ faster when $n < D = \infty$

ϕ infinite

Therefore in the Dual formulation the dominating factor is $n!$
(wrt which cost grows cubically)

kernel matrix

Φ is $n \times D$

$n \times D$

$$\Phi = \begin{pmatrix} \phi_1(x_1) & \phi_2(x_1) & \dots \\ \vdots & \vdots & \ddots \\ \phi_1(x_n) & \phi_2(x_n) & \dots \end{pmatrix}$$

$$\Phi \Phi^T = (k(x_i, x_j))_{1 \leq i, j \leq n}$$

3. $K(\mathbf{x}, \mathbf{t}) = \exp(x_1 t_1)$ where x_1 is the first component of the vector $\mathbf{x}$ and, likewise, t_1 is the first component of the vector $\mathbf{t}$
4. $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{t} - (\mathbf{x}^\top \mathbf{t})^2$
5. Now prove that if $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a given valid kernel then the following transformed kernels are also valid.
 - (a) $K(A\mathbf{x}, A\mathbf{t})$ where A is a given 2×2 matrix
 - (b) $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$ where f is a given real-valued function

Solution 1.4.1

True but result no longer in slides as [Note that for $\mathbf{c} = (c_1, c_2)^\top \in \mathbb{R}^2$, we have $\mathbf{c}^\top D \mathbf{c} = c_1^2 + 5c_2^2 \geq 0$. So, D is positive semidefinite and hence by the result on p39 of the lecture slides, K is a valid kernel. The feature map can be obtained

$$\begin{aligned}
 K(\mathbf{x}, \mathbf{t}) &= \mathbf{x}^\top D \mathbf{t} \\
 &= (x_1 \ x_2) \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} \quad \checkmark \\
 &= x_1 t_1 + \sqrt{5} x_2 t_2 \sqrt{5} \quad \checkmark \\
 &= \begin{pmatrix} x_1 \\ \sqrt{5} x_2 \end{pmatrix}^\top \begin{pmatrix} t_1 \\ \sqrt{5} t_2 \end{pmatrix} \quad \checkmark \\
 &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \quad \text{kernel } \checkmark
 \end{aligned}$$

Solution 1.4.2

Note that, for $\mathbf{x} = (1, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have 2. $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t}$ where $D := \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$

$$c^2 K(\mathbf{x}, \mathbf{x}) = (x_1 \ x_2) \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -x_1^2 + 2x_1 x_2 + 2x_1 x_2 + 4x_2^2 = \textcircled{-1} < 0. \Rightarrow \text{not a kernel}$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

Solution 1.4.3

Using the Taylor series of the exponential function $\exp(u) = \sum_{n=0}^{\infty} \frac{u^n}{n!}$, we obtain

$$\begin{aligned}
 K(\mathbf{x}, \mathbf{t}) &= \exp(x_1 t_1) \quad \checkmark \\
 &= \sum_{n=0}^{\infty} \frac{x_1^n t_1^n}{n!} \\
 &= \begin{pmatrix} 1 \\ x_1 \\ x_1^2/\sqrt{2!} \\ x_1^3/\sqrt{3!} \\ \vdots \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1 \\ t_1^2/\sqrt{2!} \\ t_1^3/\sqrt{3!} \\ \vdots \end{pmatrix} \quad \text{Taylor series expansion} \\
 &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \quad \checkmark \text{ kernel}
 \end{aligned}$$

We deduce that K is a valid kernel.

Solution 1.4.4

$$4. K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{t} - (\mathbf{x}^\top \mathbf{t})^2$$

Note that, for $\mathbf{x} = (\sqrt{2}, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have

$$c^2 K(\mathbf{x}, \mathbf{x}) = 2 - 4 = -2 < 0. \Rightarrow \text{not a kernel}$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

kernel $\leftarrow$ (a) $K(A\mathbf{x}, A\mathbf{t})$ where A is a given 2×2 matrix

Solution 1.4.5.a

(b) $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$ where f is a given real-valued function

Consider $c_1, \dots, c_m \in \mathbb{R}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^2$. Letting $\tilde{\mathbf{x}}_i := A\mathbf{x}_i \in \mathbb{R}^2$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) \geq 0 \quad \text{because } K \text{ is a kernel}$$

or equivalently

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(A\mathbf{x}_i, A\mathbf{x}_j) \geq 0 \quad \leftarrow$$

This proves that $K(A\mathbf{x}, A\mathbf{t})$ is positive semidefinite and hence it is a valid kernel.

Solution 1.4.5.b

Consider $c_1, \dots, c_m \in \mathbb{R}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^2$. Let $\tilde{c}_i := f(\mathbf{x}_i)c_i \in \mathbb{R}$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m \tilde{c}_i \tilde{c}_j K(\mathbf{x}_i, \mathbf{x}_j) \geq 0 \quad \leftarrow$$

or equivalently

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j f(\mathbf{x}_i) K(\mathbf{x}_i, \mathbf{x}_j) f(\mathbf{x}_j) \geq 0 \rightarrow \text{kernel}$$

This proves that $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$ is positive semidefinite and hence it is a valid kernel.

Problem 1.5

Consider a Gaussian kernel function $K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $K(\mathbf{x}, \mathbf{z}) := e^{-\|\mathbf{x} - \mathbf{z}\|^2}$, does there exist a finite-dimensional feature map representation? I.e., does there exist a $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ such that $K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$? Indicate an answer “yes” or “no” and provide an argument supporting your answer. [hard]

Solution 1.5

No.

Suppose for contradiction that there exist a finite-dimensional feature map representation, that is there exists $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ for some finite $d \in \mathbb{N}$ such that

$$e^{-\|\mathbf{x} - \mathbf{z}\|^2} = K(\mathbf{x}, \mathbf{z}) = \sum_{i=1}^d \phi_i(\mathbf{x}) \phi_i(\mathbf{z})$$

Consider the points

$$\mathbf{x}_\ell := (\ell, 0, \dots, 0) \in \mathbb{R}^n$$

for $\ell \in \mathbb{N} \setminus \{0\}$. For $\ell \neq h$, the angle θ between $\phi(\mathbf{x}_\ell)$ and $\phi(\mathbf{x}_h)$ in $\mathbb{R}^d$ satisfies

$$\cos \theta = \frac{\phi(\mathbf{x}_\ell)^\top \phi(\mathbf{x}_h)}{(\phi(\mathbf{x}_\ell)^\top \phi(\mathbf{x}_\ell))(\phi(\mathbf{x}_h)^\top \phi(\mathbf{x}_h))} = \frac{K(\mathbf{x}_\ell, \mathbf{x}_h)}{K(\mathbf{x}_\ell, \mathbf{x}_\ell)K(\mathbf{x}_h, \mathbf{x}_h)} = e^{-|\ell-h|} \leq e^{-1}$$

Since $\arccos$ is a decreasing function on $[-1, 1]$, we obtain that the angle satisfies

$$\theta \geq \arccos(e^{-1}) > 0.$$

This means that we have infinitely many vectors $\{\phi(\mathbf{x}_\ell) : \ell \in \mathbb{N} \setminus \{0\}\}$ in $\mathbb{R}^d$ with the angle between any pair of these being greater than $e^{-1} > 0$. This is a contradiction as there exist only finitely many such vectors (see next two paragraphs). Having reached a contradiction, we deduce that our hypothesis that there exists a finite-dimensional feature map was wrong. We have proved that there does not exist such a finite-dimensional feature map.

Consider a set of vectors $\mathcal{S}_\epsilon$ in $\mathbb{R}^d$, satisfying the property that the angle between any two vectors in $\mathcal{S}_\epsilon$ is greater than some positive constant $\epsilon > 0$. The length of the vectors does not influence the angles between them, hence we can assume that they are of unit lengths, so the vectors lie on the d -dimensional unit sphere. The fact that $\mathcal{S}_\epsilon$ must be finite is intuitive (at least for $d = 2$ and $d = 3$) and can be assumed, we prove it for completeness.

Consider the d -dimensional unit sphere. Since it is contained within $[-1, 1]^d$, the volume of this sphere is bounded by 2^d (i.e. the volume of $[-1, 1]^d$). Now consider d vectors $\mathbf{v}_1, \dots, \mathbf{v}_d \in \mathcal{S}_\epsilon$ lying on the unit sphere. Since the angles between any two such vectors is greater than $\epsilon > 0$, the polytope with vertices $\{\mathbf{v}_1, \dots, \mathbf{v}_d, \mathbf{0}\}$ in $\mathbb{R}^d$ must have volume greater than some positive value V_ϵ depending on ϵ . So, d vectors in $\mathcal{S}_\epsilon$ contribute at least volume V_ϵ . Using a crude upper bound, $d \left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right)$ vectors in $\mathcal{S}_\epsilon$ form at least $\left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right)$ non-overlapping polytopes constructed as above. Those contribute at least volume $\left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right) V_\epsilon$ which is larger than the upper bound 2^d on the volume of the unit sphere. This is a contradiction since by adding volumes of non-overlapping polytopes contained within the unit sphere, we obtain larger volume than the volume of the sphere itself. We deduce that $\mathcal{S}_\epsilon \leq d \left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right) < \infty$.

Problem 2.1

Argue that the vector space definition implies that every element $\mathbf{x} \in X$ has an additive inverse $-\mathbf{x} \in X$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$.

Solution 2.1

We have

$$\mathbf{0} = 0\mathbf{x} = (1 - 1)\mathbf{x} = 1\mathbf{x} + (-1)\mathbf{x} = \mathbf{x} + (-1)\mathbf{x}$$

where we used property 7 for the first equality, properties of $\mathbb{R}$ for the second equality, property 5 for the third equality and property 7 for the last equality (properties of vector spaces defined on page 3). We deduce that the additive inverse of $\mathbf{x} \in X$ is $(-1)\mathbf{x} \in X$ which we simply write as $-\mathbf{x} \in X$.

Problem 2.2

Kernels between sets. Let X be a finite set. Define $K: 2^X \times 2^X \rightarrow \mathbb{R}$ as

$$K(A, B) := 2^{|A \cap B|} \quad \begin{matrix} \in \mathbb{R} \\ \swarrow \\ A, B \subseteq X \end{matrix}$$

where $A, B \subseteq X$. Prove that K is a kernel.

Solution 2.2

Recall that 2^X is the power set of X (i.e. the set of all subsets of X , including the empty set), and that $|2^X| = 2^{|X|}$.

To prove that K is a kernel, we construct a feature map. For a statement S which is either true or false, we use the notation

$$[S] = \begin{cases} 1 & \text{if } S \text{ is true,} \\ 0 & \text{if } S \text{ is false.} \end{cases}$$

Now, define the features

feature indexed by subset C

$$\phi_C(A) = [C \subseteq A] = \begin{cases} 1 & \text{if } C \subseteq A \\ 0 & \text{o/w} \end{cases}$$

indexed by subsets $C \in 2^X$ (as opposed to the more conventional indices $i = 1, 2, 3, \dots$). For $A, B \in 2^X$, we then have

$$\begin{aligned} \phi(A) \cdot \phi(B) &= \sum_{C \in 2^X} \phi_C(A) \phi_C(B) = \sum_{C \in 2^X} [C \subseteq A][C \subseteq B] = \sum_{C \in 2^X} [C \subseteq A \cap B] \\ &= \sum_{C \in 2^{A \cap B}} 1 + \sum_{C \in 2^X \setminus 2^{A \cap B}} 0 = |2^{A \cap B}| = 2^{|A \cap B|} = K(A, B). \end{aligned}$$

sum over all subsets $C \subseteq X$

$\phi(A) = (\phi_{C_1}(A), \dots, \phi_{C_{2^X}}(A))$

any $A \subseteq X$
any $B \subseteq X$

This proves that K is indeed a kernel.

Problem 2.3

Min kernel.

1. Argue that $\min(x, t)$ (where $x, t \in [0, \infty)$) is a kernel for "a more complicated example" on page 7. See discussion on page 36, on going from a kernel to a Hilbert space. [technical]
2. Let $M > 0$. Find a set of basis functions $\phi_\ell : [0, M] \rightarrow \mathbb{R}$ for $\ell \in \mathbb{N}$ such that

$$\min(x, t) = \sum_{\ell=0}^{\infty} \phi_\ell(x) \phi_\ell(t).$$

[technical, very difficult]

Solution 2.3.1

Consider the inner product space $\mathcal{F}$ which consists of functions from $[0, \infty)$ to $\mathbb{R}$ satisfying

1. $f(0) = 0$
2. f is absolutely continuous (hence $f(b) - f(a) = \int_a^b f'(u) du$)
3. $\int_0^\infty (f'(u))^2 du < \infty$

with the inner product

$$\langle f, g \rangle := \int_0^\infty f'(u) g'(u) du$$

for $f, g \in \mathcal{F}$.

We consider the feature map $\phi : [0, \infty) \rightarrow \mathcal{F}$ defined as $\phi(x) := \Phi_x \in \mathcal{F}$ for all $x \geq 0$, where

$$\Phi_x(u) := \begin{cases} u & \text{if } u \leq x, \\ x & \text{if } u > x, \end{cases}$$

for all $u \geq 0$. It can easily be verified that $\Phi_x \in \mathcal{F}$ for all $x \geq 0$ (i.e. that it satisfies the three conditions). Note that, when we differentiate Φ_x for some $x \geq 0$, we obtain the step-function

$$\Phi'_x(u) = \begin{cases} 1 & \text{if } u \leq x, \\ 0 & \text{if } u > x, \end{cases}$$

for all $u \geq 0$. Note that, for $x, t \in \mathbb{R}$, we have

$$\Phi'_x(u)\Phi'_t(u) = \begin{cases} 1 & \text{if } u \leq \min(x, t), \\ 0 & \text{if } u > \min(x, t), \end{cases}$$

for all $u \geq 0$. Finally, for $x, t \in \mathbb{R}$, we have

$$\langle \phi(x), \phi(t) \rangle = \langle \Phi_x, \Phi_t \rangle = \int_0^\infty \Phi'_x(u)\Phi'_t(u)du = \int_0^{\min(x, t)} 1 du + \int_{\min(x, t)}^\infty 0 du = \min(x, t)$$

This proves that $\min(x, t)$ is a kernel.

Solution 2.3.2

We work with the kernel $k(x, t) := \min(x, t)$ for $x, t \in [0, M]$. Consider the integral operator $T_k : L^2[0, M] \rightarrow L^2[0, M]$ which for $f \in L^2[0, M]$, is defined as

$$(T_k f)(x) := \int_0^M k(x, u)f(u)du.$$

Mercer's Theorem states that, $x, t \in [0, M]$, we have

$$k(x, t) = \sum_{\ell=0}^{\infty} \lambda_\ell e_\ell(x) e_\ell(t)$$

where $\{e_\ell\}_{\ell=0}^\infty$ is an *orthonormal basis* for $L^2[0, M]$ consisting of eigenfunctions of the operator T_k with associated eigenvalues $\{\lambda_\ell\}_{\ell=0}^\infty$. Note that, letting

$$\phi_\ell := \sqrt{\lambda_\ell} e_\ell \in L^2[0, M]$$

for $\ell \in \mathbb{N}$, we then have

$$\sum_{\ell=0}^{\infty} \phi_\ell(x) \phi_\ell(t) = \sum_{\ell=0}^{\infty} \sqrt{\lambda_\ell} e_\ell(x) \sqrt{\lambda_\ell} e_\ell(t) = \sum_{\ell=0}^{\infty} \lambda_\ell e_\ell(x) e_\ell(t) = k(x, t)$$

for all $x, t \in [0, M]$. Hence, if we obtain the eigenvalues and eigenfunctions of the operator T_k , we can recover the feature map of the kernel k .

We now explicitly derive the eigenpairs for the kernel $k(x, t) := \min(x, t)$ for $x, t \in [0, M]$. An eigenpair (λ, e) is by definition a scalar $\lambda \in \mathbb{R}$ and a function $e \in L^2[0, M]$ satisfying for all $x \in [0, M]$

$$\begin{aligned} \lambda e(x) &= (T_k e)(x) \\ &= \int_0^M \min(x, t) e(t) dt \\ &= \int_0^x \min(x, t) e(t) dt + \int_x^M \min(x, t) e(t) dt \\ &= \int_0^x t e(t) dt + \int_x^M x e(t) dt \end{aligned}$$

Letting $x = 0$, we get $\lambda e(0) = 0$ and deduce that $e(0) = 0$. We now differentiate both sides with respect to x using Leibniz integral rule and the Chain rule, we obtain

$$\begin{aligned}\lambda e'(x) &= \left(\frac{\partial}{\partial x} \int_0^x t e(t) dt \right) + x \frac{\partial}{\partial x} \left(\int_x^M e(t) dt \right) + \left(\frac{\partial x}{\partial x} \right) \left(\int_x^M e(t) dt \right) \\ &= x e(x) + x(-e(x)) + 1 \int_x^M e(t) dt \\ &= \int_x^M e(t) dt\end{aligned}$$

Letting $x = M$, we get $\lambda e'(M) = 0$ and deduce that $e'(M) = 0$. Differentiating with respect to x and using Leibniz integral rule one more time, we get

$$\lambda e''(x) = -e(x).$$

This is a second order differential equation which has solution

$$e(x) = c_1 \cos\left(\frac{x}{\sqrt{\lambda}}\right) + c_2 \sin\left(\frac{x}{\sqrt{\lambda}}\right)$$

for some constants $c_1, c_2 \in \mathbb{R}$. Note that $c_1 = e(0) = 0$ and

$$0 = e'(M) = \frac{c_2}{\sqrt{\lambda}} \cos\left(\frac{M}{\sqrt{\lambda}}\right)$$

which has solutions

$$\frac{M}{\sqrt{\lambda_\ell}} = \left(\ell + \frac{1}{2}\right) \pi \quad \Longleftrightarrow \quad \lambda_\ell = \frac{M^2}{\left(\ell + \frac{1}{2}\right)^2 \pi^2}$$

for $\ell \in \mathbb{N}$. We have

$$e_\ell(x) = c_2 \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right)$$

for $\ell \in \mathbb{N}$, since we want an orthonormal basis, we require

$$1 = \|e_\ell\|_2^2 = \int_0^M e_\ell(x)^2 dx = c_2^2 \int_0^M \sin^2\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right) dx = c_2^2 \frac{M}{2} \quad \Longleftrightarrow \quad c_2 = \sqrt{\frac{2}{M}}.$$

For $\ell \in \mathbb{N}$, the eigenpairs are then

$$\lambda_\ell = \frac{M^2}{\left(\ell + \frac{1}{2}\right)^2 \pi^2} \quad \text{and} \quad e_\ell(x) = \sqrt{\frac{2}{M}} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right)$$

which give the feature map

$$\phi_\ell(x) := \sqrt{\lambda_\ell} e_\ell(x) = \frac{\sqrt{2M}}{\left(\ell + \frac{1}{2}\right) \pi} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right).$$

for $x \in [0, M]$ We have proved that

$$k(x, t) = \sum_{\ell=0}^{\infty} \phi_\ell(x) \phi_\ell(t) = \sum_{\ell=0}^{\infty} \frac{2M}{\left(\ell + \frac{1}{2}\right)^2 \pi^2} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right) \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} t\right)$$

for all $x, t \in [0, M]$.

To illustrate that this is the correct answer, we plot in Figure [1](#) the partial sum (replacing $\sum_{\ell=0}^{\infty}$ with $\sum_{\ell=0}^n$) for $n = 0, 1, 2, 3, 4$ with $M = 2$ and $t = 1$.

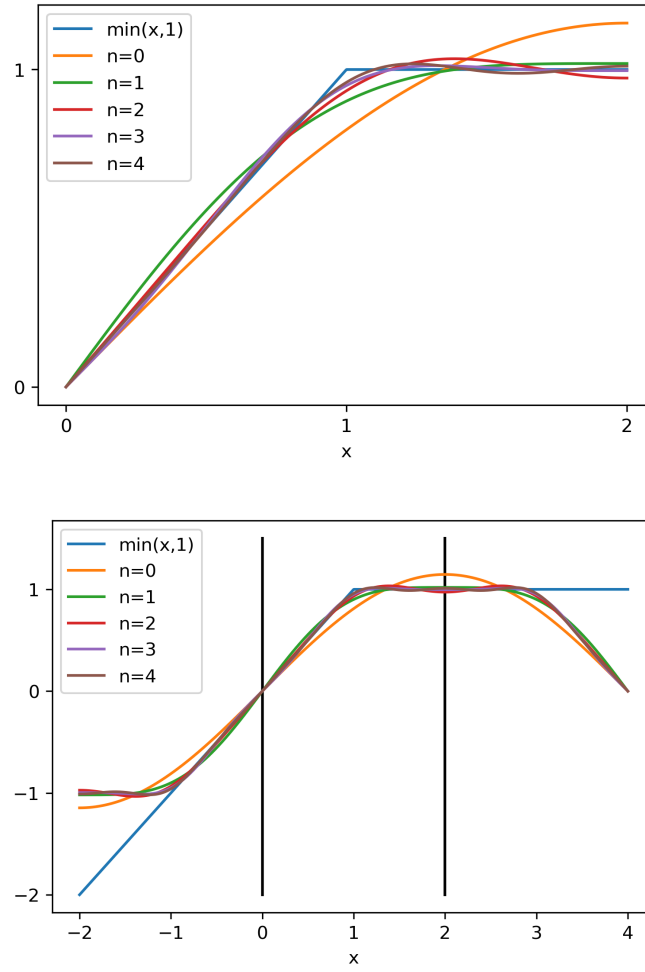


Figure 1: Min kernel partial sums.